

A protocol for *Agrobacterium* mediated transformation of *Mimulus guttatus* from leaf petiole explants

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Establishing invitro cultures for explant source (2.5-3 months):

Surface sterilization of seeds

Collect mature seeds from plants grown in greenhouse or growth chamber to reduce contamination in tissue culture. Seeds collected from natural sites/fields may increase the risks of endogenous contaminations in cultures.

1. Place seeds in a 1.5mL sterilized eppendorf tube(s) then fill the tube(s) with surface sterilization solution (5% laundry bleach and a drop of hand-soap). Shake the tube(s) vigorously for 8-10 minutes.
2. Move the tube(s) to laminar flow hood. Discard the sterilizing solution and add sterile water to rinse the seeds by shaking for at least 30 seconds. Repeat rinse at least 5-6 times to completely wash-off bleach and soap from seeds. After the last rinse add at least 1mL of sterilized water to later plate the seeds.
3. Store the seeds at 4°C for at least 2-3 weeks to cold stratify seeds.
4. Bring back tubes to laminar flow hood. Transfer the sterilized seeds onto growth medium. Spread the seeds uniformly. Grow around 5-6 seeds per jar (200mL) or polypropylene Deli jars (quart-size).
5. Incubate jars cultures at 20-21°C under cool fluorescent lamps.
6. When the plants produce proliferating axillary shoots, harvest shoots with roots and subculture onto new jar. Adjust frequency of sub-culturing to avoid senescence.

Agrobacterium mediated transformation of explants:

Agrobacterium culture preparation (4 days)

1. Streak EHA105 *Agrobacterium* strain harboring binary plasmid on LB or YEP agar plate with rifampicin (40µg/mL) and appropriate antibiotic resistance cassette on the binary plasmid. Incubate plates at 28°C for two days.
2. Inoculate a single colony of *Agrobacterium* into 10mL YEP liquid medium with rifampicin (40µg/mL) and an appropriate antibiotic (Kanamycin at 50 µg/mL or Spectinomycin at 100µg/mL or Ampicillin at 100µg/mL). Shake the culture at 28°C for 36-48 hours at 200 rpm.
3. Centrifuge the cultures to pellet *Agrobacterium* at 4000 rpm. Remove the medium and resuspend in 5mL virulence induction medium for at least 3.5-4 hours with gentle shaking (50-80rpm) in dark at room temperature. Adjusting pH to 5.5 is critical. Glucose can be substituted for sucrose, because glucose enhances *virulence* induction. However, glucose may enhance senescence of older leaves; if leaves are old, then use sucrose.
4. Adjust agrobacterium OD to 0.2 using liquid half-strength MS medium to achieve a very light color suspension in half-strength MS medium. Higher OD is not desirable.

Agrobacterium infection and co-cultivation (3-4 days)

1. Bring plant culture jar to laminar flow hood. Pull shoots onto a sterile petri dish. Using a sterile scalpel cut a leaf explant off the plant inand dip the cut petiole in agrobacterium and plate on co-cultivation medium. Perform infections in small batches to avoid desiccation of petiole explants.
2. Incubate plates in dark or low intensity light at room temperature for 2 -3 days. Avoid overgrowth of agro on plates. Overgrowth of agrobacterium may kill explants, thus the use dilute agro culture to dip petioles.
5. On day 3, wash the explants in sterile timentin (100mg/L) + cefatoxime (50mg/L) in hood and transfer the explants to callus induction medium with phosphinothricin. This is

appropriate selection medium for plasmids with *B/p* resistance cassette. Incubate plates at 21°C under cool florescent lamps.

Callus induction and shoot induction (4-5 months)

1. Subculture explants after 21-24 days. Retain only small part of callus and culture on callus induction medium to allow callus growth. It is critical to grow long enough on CPPU with routine subculture (21-24-days interval) for sufficient growth and maturation of callus. Premature transfer to Meta-toplin may kill explants. Increasing sucrose to 2.5% at later stages may help in getting larger callus. At least 3-4 subcultures are necessary.
2. When the compact callus turns friable with signatures of differentiation, pick the healthy callus and plate on shoot induction medium to regenerate shoots from callus.

Rooting of shoots (3 weeks)

1. When the shoots emerge and reach 0.5cm-1cm tall, separate individual shoots and plate them on rooting medium.
2. For good rooting and hardening, subculture on half-strength rooting medium.
3. When roots proliferate, move the shoots to potting medium in greenhouse and leave the pots on a mist bench to harden for at least two weeks.

Note:

1. Check the contents on herbicide formula to verify the active ingredient (Basta or Phosphinothricin) before use so that you can modify the selection protocol as appropriate.
2. A pilot kill-curve test may be necessary to choose the optimum concentrations of herbicide for each *M. guttatus* population.
3. Antibiotics and hormones need to be filter sterilized and added to medium after autoclaving. For best results, add antibiotics and hormones when the contents cool to 45-50°C or lower after autoclaving.
4. Fresh medium should be used for best results. Leave the solid medium overnight at room temperature after pouring into petri-dish to cool and solidify. This will minimize condensation on the lids.
5. This protocol was standardized for herbicide resistance selection.

Growth medium composition:

Murashige & Skoog basal salt medium (MS salts)	4g/L
Sucrose	20g/L
Calcium gluconate	1.3g/L
MES	0.25g/L
2-(NMorpholino) ethanesulfonic acid hydrate)	
Adjust pH to 5.6 using KOH before adding gelrite	
Gelrite	0.25%

Agrobacterium virulence induction medium:

Murashige & Skoog basal salt medium (MS salts)	2g/L
Sucrose	10g/L
MES	0.5g/L
2-(NMorpholino) ethanesulfonic acid hydrate)	
Adjust pH to 5.5 using KOH and autoclave	
Acetosyringone (Dissolved in DMSO and add before use)	200µM

Co-cultivation medium:

Murashige & Skoog basal salt medium (MS salts)	4g/L
Sucrose	20g/L

Calcium gluconate	1.3g/L
MES	0.25g/L
2-(N-Morpholino) ethanesulfonic acid hydrate)	
Adjust pH to 5.6 using KOH before adding gelrite	
Gelrite	0.25%
CPPU*	1mg/L
Acetosyringone*	100µM
* Filter sterilize before adding to the medium	

Callus induction medium:

Murashige & Skoog basal salt medium (MS salts)	4g/L
Sucrose	20g/L
Calcium gluconate	1.3g/L
MES	0.25g/L
2-(N-Morpholino)ethanesulfonic acid hydrate	
Adjust pH to 5.6 using KOH before adding gelrite	
Gelrite	0.25%
CPPU*	1mg/L
Timentin (Ticarcillin-cluvanate)*	200mg/L
Ceftaxime*	50mg/L
Phosphinothrocin*	6mg/L
* Filter sterilize before adding to the medium	

Shoot induction medium:

Murashige & Skoog basal salt medium (MS salts)	4g/L
Sucrose	20g/L
Calcium gluconate	1.3g/L
MES	0.25g/L
2-(N-Morpholino) ethanesulfonic acid hydrate)	
Adjust pH to 5.6 using KOH before adding gelrite	
Gelrite	0.25%
Meta-toplin*	0.1mg/L
Timentin (Ticarcillin-cluvanate)	200mg/L
Ceftaxime	50mg/L
Phosphinothrocin	6mg/L
* Filter sterilize before adding to the medium	

Root induction medium:

Murashige & Skoog basal salt medium	2g/L
Sucrose	10g/L
Calcium gluconate	1.3g/L
MES	0.25g/L
2-(N-Morpholino)ethanesulfonic acid hydrate	
Adjust pH to 5.6 using KOH before adding gelrite	
Gelrite	0.25%
Napthelene AceticAcid (NAA)*	0.1mg/L
Timentin (Ticarcillin-cluvanate)*	200mg/L
Ceftaxime*	50mg/L
Phosphinothrocin*	6mg/L
* Filter sterilize before adding to the medium	

Chemical supplies:

1. Timentin (GoldBio- T-104-2)
2. Cefotaxime, (GoldBio C-104-25)
3. Phosphinothricin (GoldBio P-165-250).
4. 4-CPPU Phyotech lab- C279)
5. Meta-toplin-(Phyotech lab T841)
6. Murashige & Skoog basal salts with Gamborg Vitamins (Phytotech lab: M404)
7. Acetosyringone (Sigma Aldrich Cat# 2478-38-8)